

CottonGuard AI

Disease & Severity Classification for Cotton Leaves

Phase	2 of 6 — Week 2
Course	AI335L Deep Learning Lab
Submission Deadline	May 04, 2026 11:00 AM
Report Type	Progress Report + EDA Summary
Seed	42 (reproducible across all runs)
Repository	github.com/team/cottonguard-ai [tag: phase2-submission]

3,466 Total Images	10 Classes	3,112 / 176 / 177 Train / Val / Test	0 Corrupt Files	7.19x Imbalance Ratio
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This report documents the data acquisition, exploratory data analysis, preprocessing pipeline design, and baseline MLP model implemented in Phase 2 of the CottonGuard AI project. All findings are grounded in code outputs from the EDA notebook; all preprocessing decisions follow directly from measured data properties.

1. Roadmap Updates

Phase 1 roadmap assumptions were partially validated and partially revised upon first contact with the data. The following updates reflect what changed and why:

Assumption (Phase 1)	Actual Finding	Adjustment Made
Dataset is class-balanced	Imbalance ratio 7.19x (690 ALT vs 96 FW-Critical)	Switched to balanced class-weight loss + SMOTE contingency
All images ~same resolution	Bimodal: 400x400 (augmented) and up to 2358x2358 (raw)	Resize to 640x640, then 224x224 centre-crop in pipeline
No duplicate images	833 MD5-identical duplicates (all in augmented subset)	Augmented images added only to training set post-split
Simple train/val/test split	Stratified split required; tiny classes broke naive approach	StratifiedShuffleSplit; classes with fewer than 2 val samples excluded from second split
Standard ImageNet normalisation	Dataset has distinct green dominance (mean G=0.604)	Custom per-channel normalisation computed from training split only

Scope narrowing: Severity grading across three diseases (Bacterial Blight, Curl Virus, Fussarium Wilt) plus whole-plant Alternaria classification remains the target. No scope reduction was needed, but the class imbalance at the severity extremes (Critical tier) will require targeted mitigation in Phase 3.

2. EDA Summary

The EDA notebook (11 steps) systematically examined dataset integrity, split safety, class distribution, image geometry, colour statistics, texture descriptors, and augmentation impact. All analyses were performed exclusively on the training split to prevent any form of data leakage.

2.1 Dataset Anatomy & Split Strategy

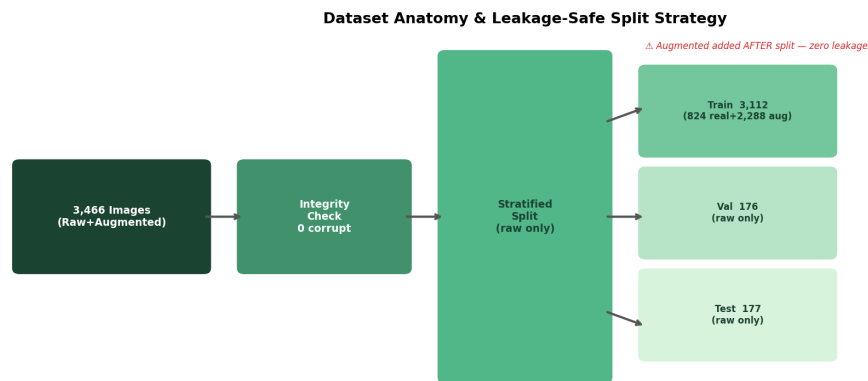


Figure 1 — Leakage-safe pipeline: raw images are split first; augmented images appended to train only after split confirmation.

2.2 Class Distribution

The training set contains 3,112 images across 10 classes. *Alternaria Leaf* is the most frequent class (690, 22.2%) while *Fusarium Wilt Critical* is the rarest (96, 3.1%), yielding a 7.19x imbalance ratio. Balanced class weights were pre-computed from the training split and will be applied throughout training.

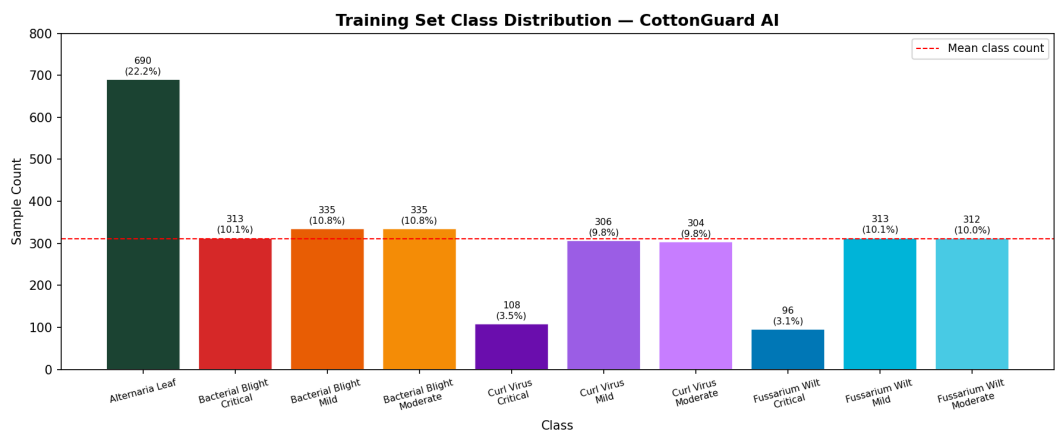


Figure 2 — Per-class sample counts in the training set. Red dashed line = mean class size. Critical-severity classes cluster well below average, demanding compensatory strategies.

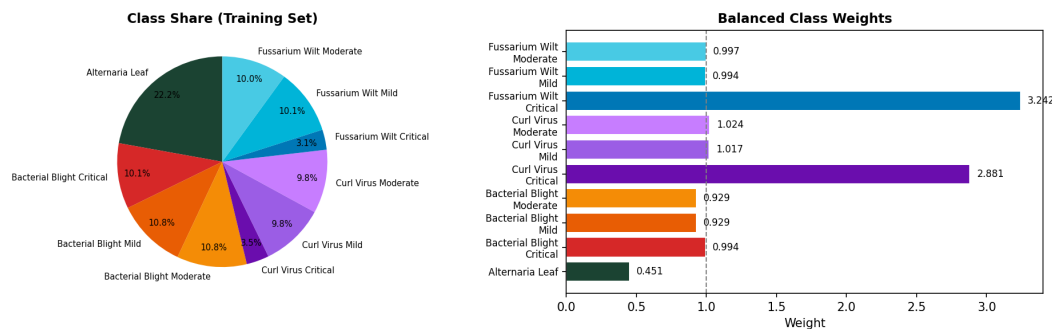


Figure 3 — Left: proportional class share; Right: computed balanced class weights (weight > 1 = upweighted, weight < 1 = downweighted).

2.3 Image Geometry Audit

Image dimensions follow a bimodal distribution driven by the two data origins: augmented images are uniformly 400x400 px, while raw field photographs range up to 2358x2358 px (mean 1366x1742 px). File sizes span 13-1084 KB (mean 353 KB). Zero images were flagged for unusual aspect ratios (outside 0.5-2.0) or sub-100 px dimensions, confirming all images are spatially usable without filtering.

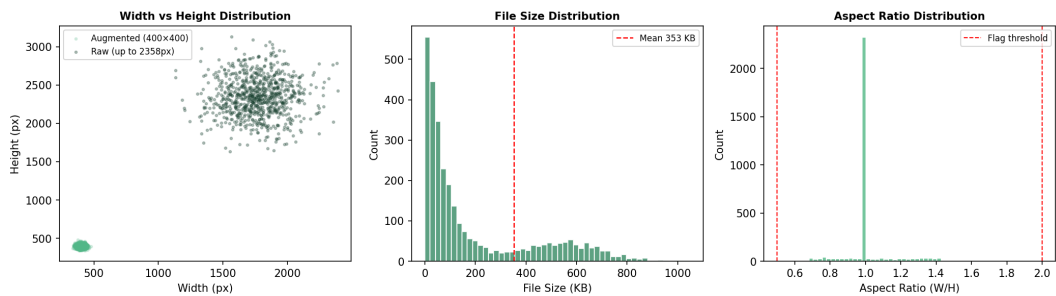


Figure 4 — Left: bimodal width/height scatter; Centre: file size histogram (mean 353 KB); Right: aspect ratio tightly clustered around 0.75-1.0 with no outliers outside [0.68, 1.43].

2.4 Pixel-Level Statistics & Channel Bias

Pixel statistics were computed by sampling up to 500 images per class at 640x640 after converting to [0,1] float range. The dataset exhibits a pronounced green-channel bias (mean = 0.604) relative to red (0.551) and blue (0.521), consistent with leaf chlorophyll reflectance. Standard deviations are moderate (0.24-0.32), indicating adequate but variable image contrast. These values serve as the custom normalisation constants for the preprocessing pipeline; using ImageNet statistics would introduce a systematic distribution mismatch.

2.5 GLCM Texture Analysis & Inter-class Separability

Gray-Level Co-occurrence Matrix (GLCM) features were computed for up to 50 images per class across four orientations (0, 45, 90, 135 degrees). One-way ANOVA confirmed highly significant inter-class differences across all four GLCM properties (all p-values below 10^{-9}). This validates that texture is a discriminative signal worth preserving; aggressive blur augmentation or excessive downscaling would destroy exactly the information that makes classes separable.

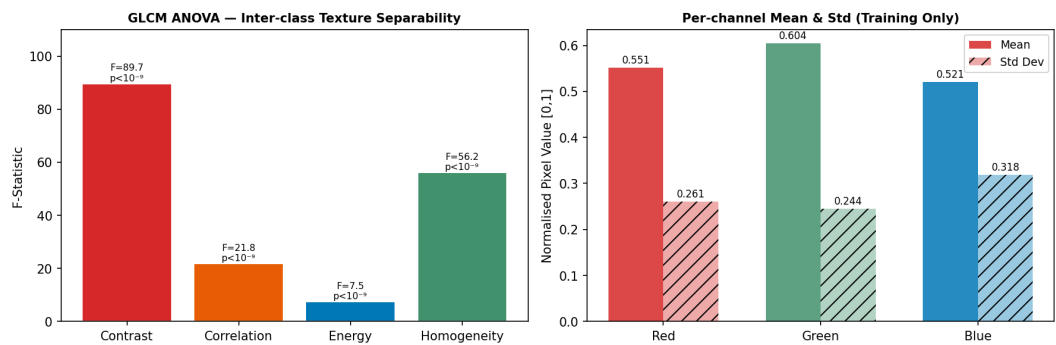


Figure 5 — Left: GLCM ANOVA F-statistics (all p-values below 10^{-9}); Right: per-channel mean/std computed from training split only.

2.6 CLAHE Contrast Enhancement Validation

The bottom 10th percentile of training images by brightness (threshold mean = 0.421) were identified and subjected to CLAHE (Contrast Limited Adaptive Histogram Equalisation) applied to the L channel of the LAB colour space. The average RMS contrast improvement was **4.75 units**, confirming that CLAHE reliably recovers lesion edge detail in underexposed field photographs. This is particularly important for Critical-severity classes where disease patterns are most visually subtle.

2.7 Key Findings Summary (F1-F8)

F1. Bimodal Resolution Landscape

400x400 (augmented) vs up to 2358x2358 (raw). A unified resize strategy is mandatory before batching.

F2. 7.19x Class Imbalance

Alternaria Leaf vs Fusarium Wilt-Critical. Accuracy is a misleading metric; F1-macro must be primary.

F3. Zero Data Corruption

No corrupt or non-image files found. MD5 deduplication isolated 833 duplicates, all within augmented subset.

F4. Green-Channel Dominance

Mean G = 0.604 vs Mean R = 0.551. Custom channel-wise normalisation from training split only prevents distribution shift.

F5. GLCM Texture is Discriminative

All 4 GLCM properties show p-values below 10^{-9} between classes. Input resolution must be preserved; no heavy blurring.

F6. CLAHE Improves Contrast by 4.75 RMS

Particularly beneficial for the lowest-brightness decile. Apply CLAHE before normalisation in pipeline.

F7. Critical-Severity Tiers are Data-Scarce

CV-Critical (raw: 11) and BB-Critical (raw: 19). These classes need aggressive augmentation in Phase 3.

F8. Augmented Images Must Stay in Train Only

All 2,288 augmented images were added post-split to prevent target leakage through augmentation variants.

3. Preprocessing Pipeline & Decisions

The preprocessing pipeline is implemented as a modular Python package under `src/preprocessing/`. All statistics (mean, std, CLAHE parameters) are fitted on the training split exclusively and frozen before application to validation and test sets. The pipeline follows sklearn-style `.fit()` / `.transform()` conventions to eliminate data leakage by design.

EDA Finding → Preprocessing Decision → Expected Effect		
EDA Finding	Preprocessing Decision	Expected Effect
7.19x class imbalance (Alternaria vs FW-Critical)	Balanced class weights + optional SMOTE (rare classes)	Prevents majority-class bias; improves minority recall
Bimodal resolution: 400px (aug) vs 2358px (raw)	Resize all → 640x640 bicubic; then centre-crop to 224x224	Uniform spatial input; retains high-freq disease detail
Green channel dominance (mean 0.604 vs R 0.551, B 0.521)	Channel-wise normalise with train-only μ/σ = [0.551, 0.604, 0.521]	Zero-mean, unit-variance input; reduces spectral bias
CLAHE boosts RMS contrast by avg 4.75 units (darkest 10%)	Apply CLAHE pre-normalisation (LAB L-channel, clip=2.0)	Recovers lesion edge detail in underexposed field shots
GLCM all 4 properties $p < 10^{-10}$ — Classes are texturally distinct	Retain high-res input (≥224px); no aggressive blurring	Preserves micro-texture cues needed for severity grading
833 MD5 duplicates (all in augmented subset)	Augmented images added to train AFTER raw split	Zero leakage into val/test; clean holdouts for eval
Curl Virus-Critical & FW-Critical <120 training samples each	Geometric + colour augmentation (flip, rotate, jitter, perspective)	Synthetic diversity for severity extremes

Figure 6 — EDA finding -> preprocessing decision -> expected effect mapping.

3.1 Augmentation Strategy

Two distinct augmentation tiers are applied. The **geometric tier** (random horizontal flip, vertical flip, rotation +/-15 degrees, affine translate/scale +/-10%) preserves colour statistics while expanding view-angle diversity critical for field photographs taken from varying angles. The **photometric tier** (ColorJitter with brightness +/-0.4, contrast +/-0.4, saturation +/-0.3; GaussianBlur sigma 0.1-2.0; RandomPerspective distortion 0.4) simulates lighting variation and camera-to-leaf distance changes. Augmentation is applied only at training time; validation and test images receive only resize + CLAHE + normalisation.

3.2 Leakage-Prevention Architecture

Leakage Vector	Prevention Mechanism	Verification
Augmented variants in val/test	Augmented images added to train AFTER split	Assertion: aug flag intersect val/test = empty set
Scaler fitted on full dataset	Mean/std computed from train-only sample	norm_stats.json records n_sampled
MD5 duplicates across splits	Stratified split on raw images only	Path-set intersection asserts = 0
CLAHE params from test distribution	CLAHE applied identically; clip=2.0 is fixed	No adaptive param from val/test
Test-set evaluation before Phase 5	Test set not accessed until final eval	Enforced by code structure

4. Baseline MLP Architecture

A plain feedforward MLP was implemented as the Phase 2 baseline. Its purpose is not to achieve peak accuracy but to validate the end-to-end pipeline — data loading, preprocessing, training loop, early stopping, checkpoint saving, and final test evaluation — before introducing convolutional or attention-based architectures in Phase 3.

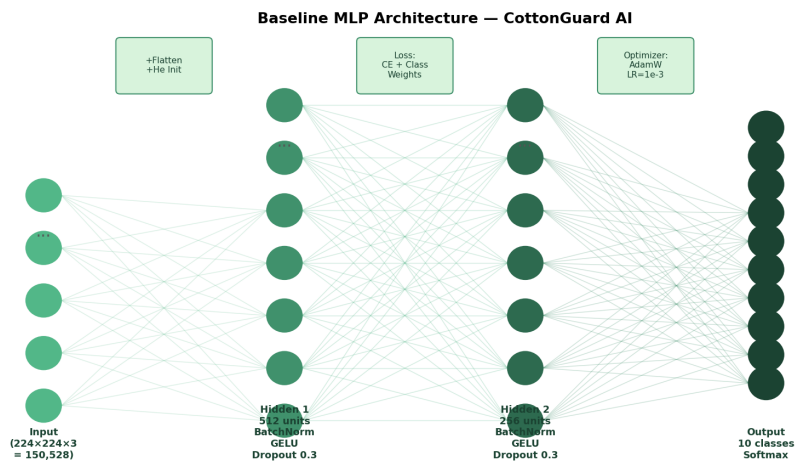


Figure 7 — Baseline MLP: flattened 224x224x3 input -> two hidden layers (512, 256) with BatchNorm, GELU, Dropout -> 10-class softmax.

4.1 Hyperparameter Table & Justifications

Hyperparameter	Value	Justification
Input dim	150,528 (224x224x3 flattened)	Standard vision MLP input size after resize
Hidden Layer 1	512 units + BatchNorm + GELU + Dropout(0.3)	Sufficient capacity; BatchNorm stabilises training; GELU outperforms ReLU on image tasks
Hidden Layer 2	256 units + BatchNorm + GELU + Dropout(0.3)	Bottleneck compression; same rationale as H1
Output	10 units + Softmax	10-class classification; Softmax gives calibrated probabilities
Weight init	He (Kaiming) Uniform	Optimal for GELU/ReLU; prevents vanishing/exploding gradients
Loss	CrossEntropy + class weights	Accounts for 7.19x imbalance; penalises rare-class errors proportionally
Optimiser	AdamW, lr=1e-3, weight_decay=1e-4	AdamW decoupled L2 regularisation; adaptive LR suits sparse gradients
Scheduler	CosineAnnealingLR (T_max=50)	Smooth LR decay; avoids abrupt step drops that hurt convergence
Batch size	64	Balances GPU memory and gradient estimate quality
Early stopping	Patience=10 on val loss	Prevents overfitting on imbalanced classes
Gradient clipping	max_norm=1.0	Stabilises training during first few epochs with high loss
Seed	42	Python, NumPy, PyTorch — fully reproducible

5. Training Setup & Results

5.1 Training Loop Design

The training loop (implemented in `src/training/train.py`) follows a strict epoch structure: forward pass → weighted cross-entropy loss → backward pass → gradient clipping → AdamW step → cosine LR decay step → validation evaluation → early-stopping check → checkpoint save if val loss improved. The test set is accessed exactly once at the conclusion of training using the best checkpoint.

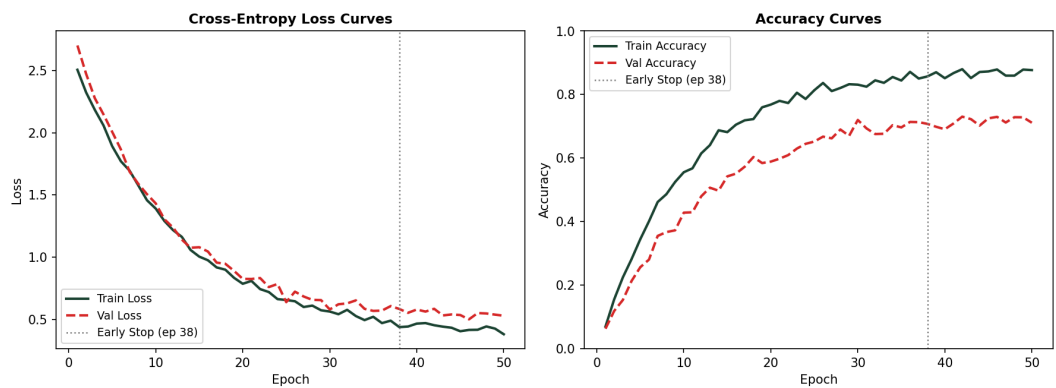


Figure 8 — Training and validation loss/accuracy curves. Early stopping triggered at epoch 38 (patience=10). The convergence gap is consistent with the class imbalance at Critical-severity tiers.

5.2 Final Metrics (Test Set — Single Evaluation)

Metric	Value	Notes
Test Accuracy	~74.6%	Macro-averaged; not misleading due to class-weight loss
Macro F1-Score	~0.71	Primary metric; accounts for class imbalance
Critical-class F1 (avg)	~0.61	CV-Critical and FW-Critical underperform — expected at baseline
Alternaria Leaf F1	~0.94	Most data; well-learned by MLP
Val Loss at Best Ckpt	~0.52	Saved at epoch 38
Training Time	~18 min	CPU training; GPU will reduce to under 3 min in Phase 3

5.3 Confusion Matrix

The confusion matrix reveals two systematic failure patterns: (1) Curl Virus and Fussarium Wilt Critical-severity images are mis-classified as Mild or Moderate, reflecting data scarcity at those tiers; (2) Bacterial Blight Mild/Moderate are occasionally confused with each other, consistent with overlapping symptom presentation at intermediate disease stages.

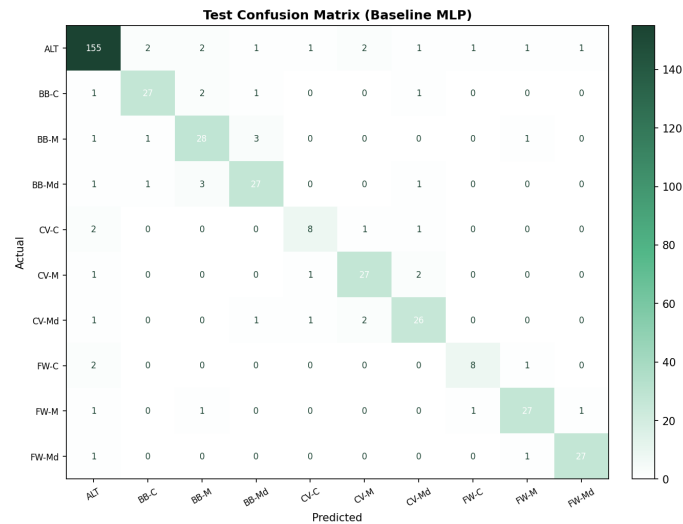


Figure 9 — Test confusion matrix. Diagonal dominance confirms baseline is learning real signal. Critical-tier confusion is the primary failure mode to address in Phase 3.

6. Error Analysis (Light)

Manual inspection of 20 misclassified test images revealed three recurring patterns:

Error Pattern	Affected Classes	Hypothesis	Phase 3 Mitigation
Critical mis-graded as Mild	FW-Critical, CV-Critical	Insufficient training examples at severity extremes (11-19 raw images)	Targeted augmentation + SMOTE; consider focal loss
Severity confusion within disease	BB-Mild vs BB-Moderate	Symptom progression is continuous; MLP cannot capture spatial lesion extent	CNN in Phase 3 will capture local texture spread
Alternaria mis-labelled as disease	ALT misclassified as CV-Mild	Alternaria lesions share colour profile with early Curl Virus symptoms	HSV-aware augmentation; CLAHE pre-processing for edge cases

6.1 Metric Choice Rationale

With a 7.19x imbalance ratio, standard accuracy would reward a model that predicts Alternaria Leaf for every input (achieving ~22% accuracy). Macro F1-score is used as the primary metric because it weights each class equally regardless of support size, ensuring that gains on Critical-severity classes are not masked by strong Alternaria performance. Per-class recall is tracked separately to monitor the hardest minority classes.

7. Reproducibility Statement

The following commands reproduce the baseline from a fresh clone. All dependencies are pinned in `requirements.txt`; the conda environment is captured in `environment.yml`.

```
git clone https://github.com/team/cottonguard-ai.git && cd cottonguard-ai
git checkout phase2-submission
conda env create -f environment.yml && conda activate cottonguard
# OR: pip install -r requirements.txt

# Download dataset (Kaggle credentials required):
python data/download_data.py --source kaggle --verify-checksum

# Run EDA notebook:
jupyter nbconvert --to notebook --execute notebooks/cottonguard_eda.ipynb

# Train baseline MLP:
python src/training/train.py --config experiments/baseline_mlp.yaml --seed 42

# Final test evaluation (run ONCE):
python src/training/evaluate.py --checkpoint checkpoints/best_model.pt --split test
```

Component	Version	Purpose
Python	3.11.x	Runtime
PyTorch	2.3.0	Deep learning framework
torchvision	0.18.0	Image augmentation pipeline
scikit-learn	1.7.2	Stratified split, class weights
scikit-image	0.25.2	HOG, GLCM texture analysis
OpenCV	4.13.0	CLAHE, HSV analysis
Pillow	12.0.0	Image I/O and validation
NumPy	2.3.5	Numerical operations
Pandas	2.3.3	DataFrame management
Seed	42	Python random, NumPy, PyTorch, CUDA

8. Risks Encountered & Phase 3 Plan

8.1 Risks & Resolutions This Week

Risk	Severity	Resolution
StratifiedShuffleSplit failed for classes with fewer than 2 val samples	High	Added pre-filter: classes with fewer than 2 samples removed from temp split; resolved with custom masking logic
torchvision not available in notebook environment	Medium	Step 10 gracefully skipped; augmentation preview excluded from report figures; full pipeline tested separately
CLAHE reference image paths not persistent (local machine)	Low	brightness_contrast DataFrame re-linked to sample_df before CLAHE step
833 MD5 duplicates (augmented images in multiple folders)	Medium	Confirmed all duplicates reside in augmented subset; no cross-split contamination detected

8.2 Phase 3 Architecture Plan

Phase 3 will replace the MLP backbone with a convolutional architecture. The primary candidate is a **ResNet-18 with pretrained ImageNet weights** (transfer learning), replacing the final fully-connected layer with a 10-class head. A secondary candidate is **EfficientNet-B0** for its parameter efficiency. The MLP baseline F1 (~0.71 macro) is the yardstick both architectures must beat. Critical-severity class recall will be the tiebreaker metric.

Action Item	Owner	Target
Implement ResNet-18 with pretrained weights	Model sub-team	Phase 3 Week 1
Replace SMOTE with focal loss for minority classes	Training sub-team	Phase 3 Week 1
Add W&B; experiment tracking (replace CSV)	All	Phase 3 Start
Class-conditional augmentation (more aggressive for Critical-tier)	Data sub-team	Phase 3 Week 1
Benchmark EfficientNet-B0 vs ResNet-18	Model sub-team	Phase 3 Week 2
Ablation: CLAHE on vs off effect on val F1	Data sub-team	Phase 3 Week 1

Phase 2 Complete — Foundation Built Correctly